

Jaziel Torres

University of Notre Dame, Department of Mathematics

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EDUCATION

University of Notre Dame

Notre Dame, IN

Ph.D. in Mathematics

Expected May 2027

Dissertation: *On Hyperbolicity Aspects of Complements of Plane Curves* (Working Title)

Advisor: Prof. Eric Riedl

University of Puerto Rico at Río Piedras

San Juan, PR

M.S. in Mathematics

May 2022

Dissertation: *Multidimensional Costas Arrays and Periodic Properties*

Advisor: Prof. Ivelisse Rubio

B.S. in Mathematics, *summa cum laude*

May 2020

RESEARCH INTERESTS

Hyperbolicity in algebraic geometry; the Green–Griffiths–Lang conjecture and its computational aspects. I am also interested in the multidimensional linear complexity of periodic arrays over finite fields, as well as the periodic properties of Costas arrays and its generalizations to multiple dimensions.

PUBLICATIONS AND PREPRINTS

- [1] Contreras, R., Cummings, J., Riedl, E., **Torres, J.** Computing Jet Differentials and the Green–Griffiths–Lang Conjecture for Complements of Smooth Plane Curves. [arXiv:2608.20479](https://arxiv.org/abs/2608.20479)
- [2] Arce, R., Hernández, C., Ortiz, J., Rubio, I., **Torres, J.** Analysis and computation of multidimensional linear complexity of periodic arrays. *Des. Codes Cryptogr.* 92, 709–722 (2024). [doi:10.1007/s10623-023-01274-w](https://doi.org/10.1007/s10623-023-01274-w)
- [3] **Torres, J.** & Rubio, I. Multidimensional Costas arrays and their periodicity. *IEEE Transactions on Information Theory*, 69(8), 5032–5040 (2023). [doi:10.1109/TIT.2023.3264951](https://doi.org/10.1109/TIT.2023.3264951)
- [4] **Torres, J.** & Rubio, I. Circular Costas maps: A multidimensional analog of circular Costas sequences. *Cryptogr. Commun.* 15, 941–958 (2023). [doi:10.1007/s12095-023-00654-2](https://doi.org/10.1007/s12095-023-00654-2)

TEACHING EXPERIENCE

University of Notre Dame

Notre Dame, IN

Instructor of Record

– Calculus II (MATH 10560) – Online

Summer 2026

– Calculus II (MATH 10560) – Online

Summer 2025

Designed and administered the online course in Canvas in the *flipped classroom* format, combining synchronous class meetings and oral exams with asynchronous instructional videos and weekly problem portfolios in which students presented solutions and reflected on their problem-solving choices.

– Calculus I (MATH 10550)

Fall 2024

Head Teaching Assistant

– Introduction to Linear Algebra and Differential Equations (MATH 20580)

Spring 2026

Coordinated the teaching assistant team in developing course-wide worksheets and led course-wide review sessions before exams.

Teaching Assistant

- Calculus I (MATH 10550) Fall 2025
- Calculus B (MATH 10360) Spring 2024
- Calculus I (MATH 10550) Fall 2023

Holy Cross College

Notre Dame, IN

Adjunct Instructor (Instructor of Record)

- Discrete Mathematics (MATH 111) Spring 2026

University of Puerto Rico at Río Piedras

San Juan, PR

Teaching Assistant

- Introduction to Computer Programming (CCOM 3030) Fall 2020

TEACHING LEADERSHIP, MENTORING, AND SERVICE

Graduate Associate, Kaneb Center for Teaching Excellence

Fall 2026 (Present)

University of Notre Dame

Facilitate workshops for graduate student instructors on teaching and course design.

Contribute to the Kaneb Center's teaching and learning blog.

Led *Foundations of Teaching*, a workshop for graduate student instructors. September 2, 2026

Mentor, Algebraic Geometry Reading Group (Volunteer)

Summer 2025

Mentored two students from the University of Puerto Rico in a reading group based on the textbook *Ideals, Varieties, and Algorithms*.

Booster Shot Review Program (Volunteer)

University of Notre Dame

Weekly targeted 30-minute review sessions in Calculus I, Calculus II, and Linear Algebra, combining focused review of key concepts with collaborative practice, and additional cumulative reviews before exams.

- **Intro to Linear Algebra and Differential Equations Coordinator** Fall 2026 (Present)
Develop review materials and oversee two undergraduate facilitators who lead the review sessions.

- **Facilitator** Fall 2025 – Spring 2026
Led sessions for Calculus I and Calculus II, and designed and facilitated the first Booster Shots for the course Introduction to Linear Algebra and Differential Equations.

TECHNICAL SKILLS

Mathematical and Computational Software: L^AT_EX, SageMath, Python, C++, Macaulay2.

Learning Management Systems: Canvas and Moodle; experience designing and administering course sites.

Mathematical Visualization Tools: Desmos, Geogebra, Manim.

Languages: English; Spanish (native).

AWARDS

Outstanding Summer Online 2026 Instructor August 2026
Notre Dame's Summer Online Program, University of Notre Dame

Outstanding Graduate Student Teaching Award April 2026
Kaneb Center for Teaching Excellence, University of Notre Dame

Richard Sady Prize October 2023
Department of Mathematics, University of Notre Dame

Awarded to the top second-year mathematics graduate student based on performance in first-year graduate courses.

Francisco Garriga Award for the Highest Distinction in Mathematics May 2020
Department of Mathematics, University of Puerto Rico at Río Piedras

Best Overall Performance and Best Performance in Geometry December 2018
Mathematics Advance Study Semester, Department of Mathematics, Pennsylvania State University

FELLOWSHIPS

Kinesis-Fernández Richards Family Fellowship 2022–Present
Graduate Fellowship for STEM Development (GFSD) 2022–Present
GEM Fellowship 2022–2023

SCHOLARLY PROGRAMS

Mathematics Advanced Study Semester Fall 2018
The Pennsylvania State University, State College, PA

Summer Program in Biostatistics and Computational Biology Summer 2018
Harvard University, Cambridge, MA. Mentor: Prof. Rafael Irizarry

Summer Internship in Biomedical Research, Informatics and Computer Science Summer 2017
University of Pittsburgh, Pittsburgh, PA. Mentor: Prof. Panos K. Chrysanthis

SELECTED TALKS AND PRESENTATIONS

Invited Talks

Computing Jet Differentials and the Green–Griffiths–Lang Conjecture for Complements of Smooth Plane Curves. Seminar Talk (Fordham University, New York, NY). September 18, 2026.

When Mathematicians Collide: The Calculus Wars and the Battle for the Foundations. Graduate Student Seminar (University of Notre Dame, Notre Dame, IN). September 26, 2025.

Entire Curves, Hyperbolicity, and Varieties of General Type: A Glimpse into the Green-Griffiths-Lang Conjecture. Graduate Student Seminar (University of Notre Dame, Notre Dame, IN). April 28, 2025.

Intro a geometría algebraica y Macaulay2: 27 líneas en una superficie cúbica [Intro to Algebraic Geometry and Macaulay2: 27 Lines on a Cubic Surface]. Seminar talk for undergraduate COMPASS scholars (University of Puerto Rico at Río Piedras, San Juan, PR). March 10, 2025.

Matemáticas, cómputos y un camino [Math, Computers, and a Path]. Undergraduate Colloquium on Mathematics and Computer Science (University of Puerto Rico at Río Piedras, San Juan, PR). February 12, 2024.

Multidimensional Costas Arrays and Their Periodic Properties. Graduate Colloquium on Mathematics (University of Puerto Rico at Río Piedras, San Juan, PR – Virtual). August 27, 2021.

My Academic Journey Through the Lens of Multidimensional Arrays. ETSU/UPR-Ponce REU Summer Program (University of Puerto Rico at Ponce, Ponce, PR – Virtual). August 6, 2021.

Linear Complexity: The Research and the Experience. Undergraduate Colloquium on Mathematics and Computer Science (University of Puerto Rico at Río Piedras, San Juan, PR – Virtual). April 21, 2021.

Contributed Talks

The Rich Geometry of Lines on Hypersurfaces. SIDIM Conference (University of Puerto Rico at Ponce). March 8, 2025.

Proofs for Two Conjectures on Multidimensional Periodic Costas Arrays. SIDIM Conference (Virtual). February 25, 2022.

Analysis and Computation of the Multidimensional Linear Complexity of Periodic Arrays. 52nd South-eastern International Conference on Combinatorics, Graph Theory and Computing (Florida Atlantic University, Boca Raton, FL – Virtual). March 10, 2021.

Multidimensional Periodic Arrays and a Measure for Complexity. SIDIM Conference (Cayey, Puerto Rico). March 7, 2020.

Poster Presentations

Computing Jet Differentials and the Green–Griffiths–Lang Conjecture for Complements of Smooth Plane Curves. ISM Discovery School: Kähler Geometry, Algebraic and Arithmetic Geometry, Hyperbolicity (Université du Québec à Montréal, Canada). August 26, 2026.